Working with data and displaying data interactively

A short course for the Department of Pharmacology and Experimental Neuroscience, UNMC , Omaha, NE

November 2026

Instructors: Dr. Mahbubul Majumder — Office: DSC 238
Phone: (402) 554-2734 — Email: mmajumder@unomaha.edu

Dr. Lochana Palayangoda — Office: DSC 235
Phone: (402) 554-3437 — Email: lpalayangoda@unomaha.edu

Lecture time: Tuesdays from 12 - 2 pm

Office Hours: After class or by appointment

Course web page: http://mamajumder.github.io/unmc26

Text books: Notes and various resources will be provided during lectures.

Recommended books:

1. Modern Data Science with R by Baumer et. al.
2. Dynamic Documents with R and Knitr by Yihui Xie

Course outline (approximate): There will be six sessions in three days. Each session will be for an hour.

Day	Session	Chapter Description	Date
1	1	Introduction and Organization of Data	Nov 09, 2026
1	2	Effective Visualization of Data	Nov 09, 2026
2	1	Supervised Learning from the Data	Nov 19, 2026
2	2	Unsupervised Learning from the Data	Nov 19, 2026
3	1	Creating Reproducible Reports Using R	Dec 07, 2026
4	1	Demonstration of a Case Study with Experimental Neuroscience Data	Dec 17, 2026

Practical Assignments: There will be six practical assignments throughout the course.
Course outcomes: At the end of the course, participants will create a data analysis report.